now familiar with them. Graphs could be considered a generalization of tree, although in reality the opposite is true, and it is trees that are a special case of graphs: a tree, in fact, is a connected, acyclic undirected graph. Figure 14.1 shows an example of two graphs, only one of which is a tree: don't worry if you are not sure which one, in this chapter we'll take a closer look at the meaning of those properties in the definition of trees, and they will help us explain graphs' properties better, and make sense of these examples.

But to do so we need to follow a meaningful order and so, before that, we need to give a formal definition of graphs, and understand how we can represent them.

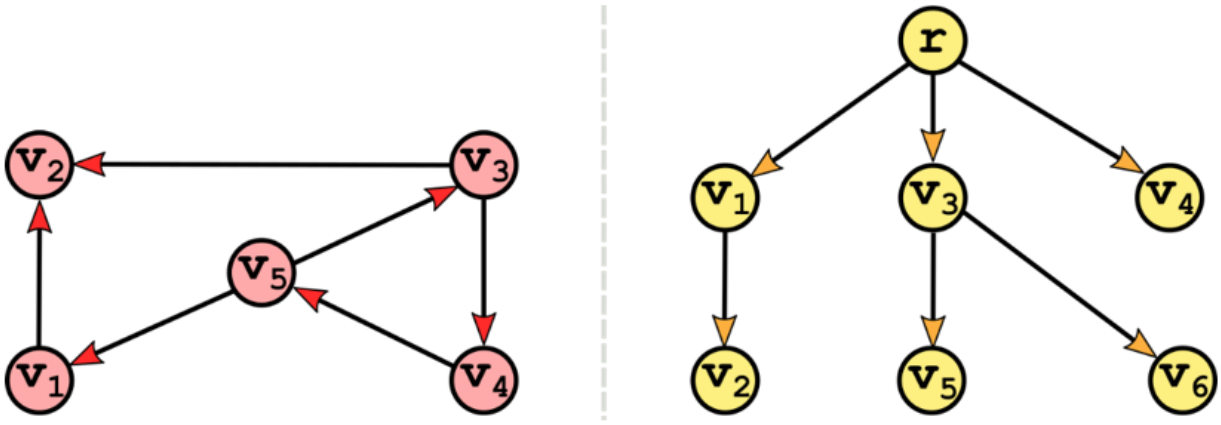


Figure 14.1 Two graphs, only one of which is also a tree: can you tell which one?

Once we have laid that foundation, we can start modeling interesting problems with graphs and developing algorithms to solve them.

In particular, here we will focus on the "shortest distance" problem; in chapters 8-11 we have developed our example of an e-commerce platform: we have introduced k-d trees